

**SIMATS ENGINEERING
(SAVEETHA SCHOOL OF
ENGINEERING) Department of
Computer Science and Engineering
ASSESSMENT TOOL 2- Short Answer**

Course Code:

CO Assessed:

Weightage:

ITA03

CO1

30% of CIA

CO1 Statement:

Student Name:

P. Premasree

Section / Batch:

Code of Conduct

Course Title:

Mobile computing

Assessment:

Total Marks:

Assessment Tool 2 - Short

Answer 100Marks

*Analyze the mobile network components, mobile base station,
mobile **infrastructures** and protocols in.
cellular systems.*

2023-2027.

(Final year)

Reg. No.:

Date:

192373006

23/07/2026

17. Bomasee (192.37206) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will*

result in disciplinary action.

Signature of the Student: P

Pus

Instructions

This test contains 50 questions, Total: **100**

Marks.

Duration: 1 hour.

98

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a

Mobile
Computing

=

Full-duplex = 2

channels

voice channels

Operator

Test

$45\text{MHZ}=45000\text{ KHz}$

Voice channels

$$\begin{aligned} &45000 \\ &= \\ &\quad 900 \\ &\quad 45006 \\ &\quad 59 \\ &\quad 2 \times 25 \\ &= 900 \end{aligned}$$

P-Premasse
e

192373006

21 Voice channels = 560

cluster = 7-cell frequency

Capacity = 560-80
charlets

3)

Carriers

=100

7

eat carrying & time
slots

Voice channels = 100×8

=800

per
hour!

Power gap = 10

$\log_{10} (39) = 10$

$\log (20)$

1008

10

(20)

£13.01dB

$$-60 \text{ dBm} - (-92) \text{ dBm} = 60 + 92$$

5)

Ratio((/ I
)

$$= 60 + 92 = 32 \text{ dB}$$

6)

Spectrum = BookHz 42MHZ

42000

$$= 140 = \underline{140} = 20 \text{ channel}$$

42MHZ 300KHZ

300

7

∞

10

$$\Rightarrow \text{channels} = 140$$

$$= 20$$

7

Received

power

64

= 84W

24

Power = 22

=4 times

total channels = 15x4=66

Doppler
shift

100km/hr =

27-78m/s

$FD = (V/C)$

f

Doppler shift = 83

H 2

11)

duration=

$$15 = 0 - 625$$

.

12/ Carrier frequencies = 24
MHz

MHz

200 KHz

13) Here 8W = 39
dB

=

=) Power = 39dBm -
33dBm

= 6dBm

FSPL = $32.44 + 20 \log$
 $(1900) + 20 \log$

Total capacity =

$6 \times 250 = 1500$ cells

14)

15)

$$120 \times 24006$$

=120 carriers

$$208$$

$$(2-5) = 106 \text{ dB } (2-5)=106$$

$$3 \text{ min} = 180 \text{ s}$$

No- of active calls 4000

$$=22 \text{ call}$$

$$180$$

$$17) \text{ Total bandwidth} = 150 \times 25 \text{ kHz} = 3750$$

$$-3.75 \text{ MHz}$$

$$18$$

Total

$$\text{bandwidth} = (30-1) \times 40 = 1160 \text{ kHz}$$

$$2 = 1.16 \text{ M} / +12$$

19

90

20)

22

$$+9 \rfloor -3 \circ \rfloor$$

1.5×2

. ∴ handoffs=30

mar

Total capacity= $8 \times 40^{*} 250$

$$= 80,000$$

$$\text{No of cells} = (0.4)^2 = 0.16400$$

$$\text{channels} = 6000 = 200$$

30

2500

0-16

23

total

$$1200 \times 4 \times 2-5$$

traffic=

60

$$= 200$$

Extangs

$$24) \text{Max users} = 40000 = 100$$

Total capacity

про

25 Pasily

$$\text{bath} = 600 \times 1200 = 7200$$

26) Paging rounds =

$$200 - 50$$

e
20
te
une

001

2471 offered
traffic

$$= 40000 \times 3 = 120000$$

Blocked =

$$(20,000 / 120000 \times 100 =$$

16.7% blocked

! No of subscribers -15% of
600000-90,000

28)

$$29) \text{ / capacity} = \frac{9000}{12000} \times 100 = 75\%$$

9000

$\times 100 =$

75%

utilization &

$$30) \text{ Total channels} = \frac{48000}{200} = 240$$

200

$- 2840$

Per cell = 240 channels.

$= 60$

4

3,

No. of

so

chargers

32

33

34

35)

36

37)

"Thannels: 30000 — —
200=150

No. of Calls: 2500 x99% = 2475
handoffs.

12W=40-8&BM =)

40-8-45=-4-2 dBm

7x900 = 6300 wers

4-8-8-0-6ms

ng

$$V = 150 \text{ km/h}$$

$$= 41.67 \text{ m/s}$$

$$f_D = (V/c) +$$

$$= 25012$$

$$\text{offered} = 70000 \times 4 = 280000$$

$$\text{Capacity} =$$

$$250000$$

Blocked,

$$\approx 30000$$

$$280000$$

$$9 \times 600 = 5400 \text{ users}$$

$$\times 100 = 10$$

40)

$$60000 \times 0.4\% = 240 \text{ calls}$$

/ Rec dropped

$$\text{loss} = 43 - (-87) = 130 \text{ dB}$$

path loss

$$41) \quad 2 - 5 \times 150 = 375$$

km?

42) Total channels -
48000

43)

49)

45)

46)

Per cell

200

=240

6

=

150000

6000

नल

$$=240$$

40

channels

५०

=25 paging
cycles

=3 handoffs

$$(-85) - (\sim 115) =$$

20 dB SNR

$$250 \times 150 \text{ kbps} = 37500 \text{ kbps} =$$

37.5Mbps

$$1500 \times 4 \times 2^{-5} / 60 = 250$$

$$2.5 / 60 = 250$$

Eslangs

47)

48)

$$250 \times 12 \text{ KHz} = 3000 \text{ KHz} \geq 3 \text{ MH}$$

49)

$$1500 \times 600$$

kbps

50) Tower area $= \pi r^2 \times 4 = 50.27$

km²

$$\frac{350}{50.27}$$

$$= 6.96$$

~7 towers

$$\text{kbps} = 900,000 \text{ Kbps} =$$

$$900 \text{ MBps}$$